Journal of Autism and Developmental Disorders
https://doi.org/10.1007/s10803-025-06920-x

ORIGINAL ARTICLE

Effectiveness of a Peer-Mediated Intervention During iPad-Based Academic Tasks on Social Behaviors of Four Students with Autism Spectrum Disorder

Gaëtan Briet[1] · Gaïd Le Maner-Idrissi[2] · Tanguy Seveno[3] · Sandrine Le Sourn-Bissaoui[2]

Accepted: 2 June 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract
The present study aimed to examine the effectiveness of a peer-mediated pivotal response treatment implemented during iPad-based academic tasks on the social behaviors of children with autism spectrum disorder (ASD) and typically developing (TD) peers. Four children with ASD and four TD children attending mainstream school settings participated. A multiple baseline design across dyads was used to measure the effects of the intervention on the social behaviors of participants. After the intervention, all children increased their rates of social responses, one child with ASD increased his rate of social initiations and three TD children increased their rates of social initiations. Additionally, inappropriate social behaviors decreased for three children with ASD. The improvements were maintained for all participants. These findings suggest that a PMI-PRT implemented during iPad-based academic tasks may be a viable option for targeting the inclusive education of students with ASD, especially for promoting positive social relationships with peers.

Keywords Peer-mediated intervention · Autism spectrum disorder · Technology · Social behaviors · Inclusive education

Introduction

Since the Salamanca statement in 1994, numerous international organizations have called for the development of inclusive practices and school inclusion has become a priority in many countries worldwide (Kurth et al., 2018). The goal is to give children with disabilities, including those with autism spectrum disorder (ASD), the opportunity to study with their typically developing (TD) peers in a non-restrictive environment that adapts to their special needs (Guillemot et al., 2022). However, the deficits in social communication skills of children with ASD often limit their reciprocal exchanges with TD children and are a major source of isolation in inclusive settings (Gunning et al., 2019). Indeed, ASD-related features are often stigmatized by TD students, who tend to have a more negative attitude and behavioral intentions toward children with ASD in comparison to TD children (Aubé et al., 2021). Consequently, students with ASD are generally less well-accepted by their classmates, and experiences of bullying at school are very frequent (Park et al., 2020). Thus, it is important to introduce interventions facilitating positive social interactions between students to improve the social participation and acceptance of children with ASD in school settings (Camargo et al., 2016), and to maximize the potential benefits of inclusive education on their development (Briet et al., 2021).

Peer-mediated intervention (PMI) is an evidence-based method for promoting the social inclusion of students with ASD (Crosland & Dunlap, 2012). In its broader conception, PMI refers to an approach in which a child or a group of children are taught to use instructional techniques (e.g., prompting, tutoring) to produce changes in a target child's social, communication, and/or academic behaviors (Odom & Strain, 1984). In ASD research, PMI often teaches TD children to help children with ASD actively participate in social interactions, thereby enabling children with ASD to develop appropriate social skills (Zhang et al., 2022). Numerous studies have shown that PMI increases a broad range of skills in students with ASD, such as joint attention (Hansen et al.,

✉ Gaëtan Briet
gaetan.briet@unimes.fr

[1] NÎMES UNIV., APSY-V, 30021 Nîmes Cedex 1, France

[2] Université Rennes, LP3C (Laboratoire de Psychologie, Cognition, Comportement, Communication), Rennes, France

[3] Centre Hospitalier Guillaume Regnier, Pôle I.O3, Sessad Mille Sabords, Rennes, France

Published online: 22 June 2025

2023), social initiation and social response (Hu et al., 2018), play behaviors (Kent et al., 2021), and conversational abilities (Thomas & Bambara, 2020). Several systematic reviews and meta-analyses also demonstrated the efficacy of PMI in increasing positive social interactions between children with ASD and their TD peers in inclusive settings (Gunning et al., 2019; Watkins et al., 2015; Whalon et al., 2015).

Among the existing peer-mediated programs within the literature, the peer-mediated pivotal response treatment (PM-PRT; Boudreau et al., 2015; Koegel et al., 2001) seems promising for improving the social skills of children with ASD at school. The PM-PRT is a naturalistic behavioral intervention targeting pivotal areas (such as motivation for learning and communication), in which changes produce collateral gains in children's behavior (Boudreau et al., 2015; Koegel et al., 1999). To date, a series of studies conducted in school settings demonstrated that the PM-PRT is a potentially effective method to address the social needs of students with ASD (Boudreau et al., 2021; Brock et al., 2018; Harper et al., 2008; Kuhn et al., 2008; Pierce & Schreibman, 1995), while highlighting the need for further research. Notably, the PM-PRT, as well as PMI as a whole, has been mainly implemented in a play context (Boudreau et al., 2015), and very few peer-mediated studies have targeted participants during academic activities (e.g., Kamps et al., 1994; Tan & Alant, 2016). Yet, school activities are an important part of students' school day, as well as an additional opportunity to foster mutual support and relationships between children with disabilities and their peers (Moeyaert et al., 2021). Thus, because the PM-PRT is designed to be conducted in the context of everyday routines (Boudreau et al., 2015), it seems important to examine whether such an intervention could be implemented during academic tasks in school settings. Furthermore, according to the socio-constructivist approach (Bruner, 1983; Vygotsky, 1978), effective social interactions are necessary for learning. Consequently, increasing positive social exchanges between children with ASD and their TD classmates during academic tasks may also promote academic learning in children with ASD. Empirical data also support this idea, demonstrating that PMI can lead to positive changes in social and academic behaviors (Haas et al., 2019).

Over the past 10 years, it has also been argued that innovative technologies could help support the inclusive education of children with ASD (Fage et al., 2018). Specifically, some studies demonstrated that the use of a handheld electronic device such as an iPad increases the motivation and academic engagement of children with ASD when compared with traditional support (Neely et al., 2013). Other studies showed that technology-based interventions may be effective for improving on-task and task-completion behaviors in students with ASD (Beckman et al., 2019; Rosenbloom et al., 2019). When shared with other students, touchscreen tablets can also encourage positive social interactions between participants (Gal et al., 2016; Paquet et al., 2022). Moreover, a few studies have recently demonstrated that PMI could be associated with a technology-based intervention to increase the social communication skills of children with ASD. For example, Thiemann-Bourque et al. (2018) investigated the impact of incorporating a PMI into a speech-generating device intervention (iPad with voice output app) on the communication of preschoolers with ASD and TD peers. Participants who received the treatment increased their rates of spontaneous communication acts more than children in the comparison group while being able to maintain their gains and generalize improvements. Dueñas et al. (2021) evaluated the effects of a multicomponent PMI incorporating video-modeling and peer self-management strategies on social exchanges between children with ASD and TD children. Results showed that all three participating dyads improved after the introduction of the intervention, although the observed effect for social response generalization of children with ASD was weak. These results are promising but additional research is needed on the use of technology-based instruction in PMI approaches for students with ASD.

In summary, most previous research on PM-PRT has been implemented in play contexts (Boudreau et al., 2015), and there is a lack of studies evaluating the effectiveness of PMI during academic activities (Tan & Alant, 2016). Moreover, only a few studies combined PMI and technology-based instruction (Dueñas et al., 2021; Thiemann-Bourque et al., 2018), although the use of iPads/tablets appears promising to increase the academic engagement and social skills of students with ASD (Gal et al., 2016; Neely et al., 2013; Rosenbloom et al., 2019). To address these gaps, this study aimed to explore the effectiveness of a PMI adapted from the PM-PRT during academic tasks on an iPad in four students with ASD attending ordinary school settings. We adopted a single-case multiple baseline design and asked whether such an intervention could improve participants' appropriate social behaviors and reduce their negative social behaviors. Specifically, we addressed the following research questions:

1. Was the intervention effective in improving the participant's positive initiations and responses?
2. Was the intervention effective in decreasing participant's negative initiations and responses?
3. Were the effects of the intervention maintained in the short term when the adult's support was withdrawn?

Method

Participants

Four children with ASD and four TD children from two schools in France participated in the study. Both schools had

a self-contained classroom for students with ASD that was affiliated with the university. To be included in the study, children with ASD had to be enrolled in one of these two classes and meet the following criteria: (a) having a formal diagnosis of ASD established by a child psychiatrist using the Autism Diagnostic Interview, Revised (ADI-R; Rutter et al., 2003) and the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2; Lord et al., 2012), (b) being familiar with the use of a portable touch-screen tablet, (c) lacking social initiation and response (i.e., exhibiting consistent failure to independently start and appropriately react to social interactions, as measured by specific behaviors like not greeting others, ignoring their name, or failing to engage in reciprocal communication or play with peers), and (d) needing substantial support (i.e., gestural or physical prompting) to socially interact with TD peers when the opportunity arose. To be included in the study, TD children had to attend an ordinary class that shared at least one activity during the day (e.g., lunch, recess) with the class of children with ASD, to ensure that participants were already familiar with each other before the beginning of the study. Then, as recommended in PMI studies (Charlop et al., 2018), TD children were nominated for participation by their teacher based on the following criteria: (a) being a volunteer to participate in the sessions, (b) attending school regularly, (c) being receptive to instructions from an adult, (d) exhibiting age-appropriate language and social skills, and e) being academically able to compensate for absences from class during the sessions. A more detailed description of the participants is presented below.

Arthur was a 4;5-year-old boy, diagnosed with ASD and global developmental delay. His overall developmental age was estimated at 2;3 years and his receptive language was estimated at 2 years with the Psychoeducational Profile-Third Edition (PEP-3; Schopler et al., 2005). He understood sentences of three words and was able to follow one-step instructions (e.g., give me the toy). In terms of expressive language, Arthur was non-verbal and communicated exclusively through a speech-generating device based on the Picture Exchange Communication System (PECS; Bondy & Frost, 1994). During recess, his social participation was minimal. He frequently engaged in self-stimulatory behaviors, such as fixating on lines on the ground or repeatedly producing non-communicative sounds, without interacting with peers or engaging in structured activities. With adult prompting, he was able to share a common space and play alongside other children but expressed difficulties interacting with them. Arthur was paired with a 5th-grade male student attending an ordinary class that shared lunch hour with Arthur's class.

Bastien was a 10-year-old boy, with a diagnosis of ASD and a global developmental delay. His results from the PEP-3 indicated an overall development estimated at 4;5 years, a receptive language development estimated at 4;8 years, and an expressive language development estimated at 4;4 years. He was able to understand and follow two-step instructions (e.g., call a peer and show him your puzzle), and he communicated in three-to-six-word sentences. He was also able to initiate brief conversations with an adult to make requests, ask questions, and discuss topics of interest. However, social interactions with peers were much rarer. During recess, he generally remained isolated, observing other children or displaying self-stimulation. When other children came up to him, he often remained unresponsive or exhibited aggressive behaviors, such as pushing them away. Bastien was paired with a 5th-grade male student, attending an ordinary class adjacent to Bastien's. The two classes shared the same daily recess time. In addition, once a week for around 45 min, Bastien was included in this ordinary class.

Carl was a 8;8-year-old boy, diagnosed with ASD and global developmental delay, as indicated by an overall developmental age estimated at 2;1 years with the PEP-3. He was nonverbal and his receptive language development was estimated at 1;10 years with the PEP-3. Carl could vocalize to draw attention or express refusal but essentially used the PECS to communicate his needs and make requests. He understood around ten isolated words and sentences of two familiar words. Socially, he was able to imitate two-step actions and participate in social play with an adult. During recess, he observed the other children and could engage in parallel play with adult prompting, but rarely spontaneously and showed limited endurance in reciprocal interactions. Carl was paired with a 5th-grade male student attending an ordinary class that shared daily playtime with Carl's class. Once a week for 30 min, Carl was also included in his peer's class.

David was a 6;8-year-old-boy, with a diagnosis of ASD and intellectual disability. His intellectual development was estimated at three to four standard deviations below the norm, as reflected in an IQ score of 50 on the Snijders-Oomen Nonverbal Intelligence Test (Tellegen et al., 2009). He was nonverbal and his receptive language development was estimated at 1;9 years with the PEP-3. He used the PECS as a primary means of communication (essentially to make simple requests), although he was also able to initiate joint attention by pointing at objects to indicate his wish. He understood fewer than 10 familiar words and followed one-word instructions. During recess, he frequently walked around the playground, engaging in self-stimulatory behaviors (e.g., rubbing his fingers near his face), and did not participate in structured activities. With adult prompting, he occasionally shared a play area with other children (e.g., a slide), but always for a very short time and without direct interactions with his peers. David was paired with a 5th-grade male student attending an ordinary class that shared daily playtime with David's class.

Settings

Participants with ASD attended a self-contained special education class at an ordinary school complex in Western France. Each class served a total of seven to nine children with ASD and was staffed by a multidisciplinary team including a special education teacher and several practitioners (educators, psychologists, motor therapists, speech therapists, and occupational therapists). In each school, study sessions took place in a quiet room in the school, known to all participants as it was frequently used for other school activities. The room contained one table with three chairs, and cupboards containing various materials for daily activities. All sessions were recorded for measurements using a Sony® HDR-CX405 camcorder on a tripod and placed facing the children.

Materials

The materials included an Apple iPad with the Preschool® app (Bourgueil et al., 2015). We chose to use this application for several reasons. Firstly, it is an application that has been specifically designed for children with ASD, incorporating learning strategies that consider their special needs (e.g., limiting distractors, reinforcing correct answers). Secondly, it is one of the few applications available in French that includes academic activities covering a wide spectrum of developmental ages (from 1 to 6 years), making it suitable for all the children in the present study. Preschool® offers a wide range of academic activities (e.g., literacy, numeracy, color identification, concept categorization), graded according to their level of difficulty. For the present study, four activities were selected for each child. All activities were selected by the participants' teachers, based on each student's academic skills and educational program. Consequently, during all the study sessions, Arthur and his peer carried out puzzle-solving, picture categorization, color matching, and letter drawing activities. Bastien and his peer carried out puzzle-solving, picture categorization, double-entry switchboard, and visual discrimination activities. Finally, Carl, David and their peers carried out puzzle-solving, picture categorization, letter drawing, and visual algorithm activities. Figure 1 illustrates the interface of the Preschool® app during a puzzle-solving activity.

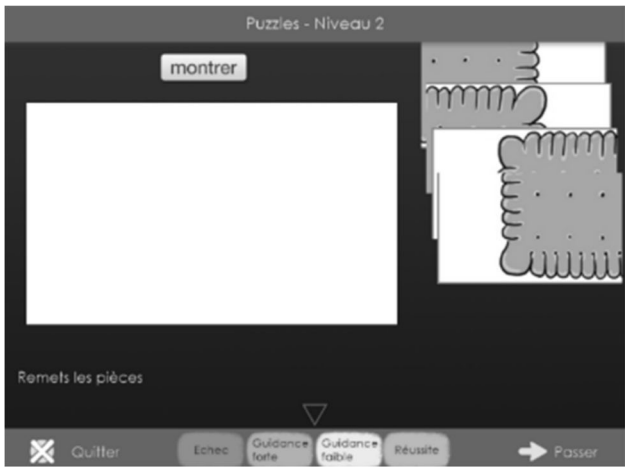


Fig. 1 Example of a puzzle-solving activity in the Preschool® app

Experimental Design

A single-case multiple baseline across participants design (Horner et al., 2005; Krasny-Pacini & Evans, 2018) was used to assess the effects of the PM-PRT on the social behaviors of participants. Because the intervention could not start at the same time for all participants, the present study incorporated non-concurrent multiple baselines (Christ, 2007; Watson & Workman, 1981), which is an extension of a traditional multiple-baseline design particularly suited to applied research in multiple educational contexts such as ours (e.g., Alzrayer et al., 2019; Briet et al., 2023). In a non-concurrent multiple baseline design, participants are randomly assigned to predetermined lengths of baselines, with the goal of demonstrating changes in the dependent variables only when the intervention is introduced (Christ, 2007; Watson & Workman, 1981). According to these standards, participants in the present study were randomly assigned to baseline lengths of six, seven or eight sessions. Then, the intervention was delivered for ten sessions (only nine for David due to repeated absences) and the dependent variables continued to be assessed over four follow-up sessions (except for David who was unable to complete this phase).

Dependent Variables

All data were scored and graphed from video recordings of the study sessions. The first author was the primary observer and coded dependent variables using The Observer XT software (Noldus Information Technologies). Partly based on the research of Katz and Girolametto (2015), four dependent variables were defined and assessed for all participants: initiations, negative initiations, responses and negative responses. Each dependent variable was coded by counting the number of behaviors per session that we then converted to a rate per minute. Table 1 provides a definition and specific examples for each dependent variable.

Procedures

General Procedures

This study was conducted in accordance with the terms of the Declaration of Helsinki regarding ethical principles applicable to research involving human beings, as well as with the French legislation regarding psychological experiments. Written consent was obtained from the parents of all children included. After age-appropriate explanations, oral consent was also obtained from all TD students and one child with ASD. Since explicit consent was not possible for the other three children with ASD who were nonverbal, the experimenters paid particular attention to non-verbal indicators of their well-being and stopped the sessions in the event of significant discomfort.

Thirty-minute sessions were conducted at fixed days and times for each target child, two times per week for approximately six months. In each dyad, the experimenter was the same throughout the study. To create routines and rituals for participants with ASD, a clear timeline was also established within all sessions, as recommended in some well-validated peer-mediation models (e.g., Wolfberg et al., 2012): say hello and sing a song (~ 5 min), work together on the iPad (~ 20 min), put the tablet away, sing a song and say goodbye (~ 5 min).

TD children Pre-training on the iPad

At the request of school staff, the experimenters completed pre-training with the TD students. They were taught to use the iPad and to complete the tasks from the Preschool® app. They could also ask the experimenter questions about the sessions and what was expected of them. When the children were able to use the application independently (i.e., without adult support), the pre-training on the iPad was considered complete. For all TD participants, this phase lasted no longer than one 30-min session.

Baseline

During baseline, participants were instructed to perform the four tablet-based academic tasks previously selected. The TD participant was also encouraged to support his peer with ASD without any prompt, guidance or feedback from the adult. The experimenter was seated in a corner of the room but interacted as little as possible with the participants. He occasionally solved a problem with the iPad and ensured that the children were safe during the sessions.

Table 1 Definitions of dependent variables and examples of behaviors coded across the dyads

Variable	Definition	Examples
Initiation	A peer-oriented behavior (gesture and/or verbalization), which began an interaction, was not prompted by the peer's behavior and was positive in nature (i.e., affiliative or prosocial)	Ask for the tablet; offer help; make a comment; try to get the other's attention; congratulate the peer
Negative initiation	A peer-oriented action (verbal or physical), that began an interaction, was not prompted by the peer's behavior, and was uncomplimentary, rejecting or physically harmful in nature	Try to snatch the tablet out of the other's hands; climb on the other child; cry toward the peer; say "You suck"
Response	A behavior (eye-gaze, gesture and/or verbalization) that was directed toward the peer or a referent named by the peer (e.g., the iPad), positive in nature (i.e., affiliative or prosocial), and that occurred within an interval of 3 s of an initiation or a response	Accept a high five; look at the peer who is calling him; imitate the other; answer a question; move the puzzle piece to where the peer is pointing
Negative response	An action (verbal or physical) oriented toward the peer or a referent named by the peer, that was uncomplimentary, rejecting or physically harmful in nature, and that occurred within an interval of 3 s of an initiation or a response	Push the other's hand away; cry, strike or bite the peer who asks for the tablet; throw objects following a support proposal

TD Children Training

The peer training took place over four 30-min sessions. Participants with ASD were not present during these sessions, and the TD children were trained individually by a qualified experimenter (a licensed psychologist preparing a PhD in developmental psychology or a specialized educator with extensive experience). Both experimenters were trained in ASD and PMI-PRT. The first session consisted in reading and discussing the children's book "Epsilon: un enfant extraordinaire" (Laurent & Cellier, 2013) and in watching the animated film "Mon ami Tom" (available online at https://www.youtube.com/watch?v=-eHtZHH1AYQ). Both describe the characteristics of a child with ASD (strengths, difficulties, and needs), from his own perspective in the film and from his friend's perspective in the book. The goal was to inform TD students about ASD and related needs, using age-appropriate vocabulary. Then, the experimenter led the discussions about the importance of accepting differences between individuals and introduced the "peer buddy" approach.

The remainder of the intervention procedure included a formal peer training process consisting of verbal explanations of behavioral strategies, modeling, role-playing, and feedback, which is a widely used and effective approach in the PMI literature (Watkins et al., 2015). Thus, during the second session, the TD children were introduced to seven teaching strategies that are core components of pivotal response training (Koegel et al., 1999). These strategies were adapted from the original study conducted by Pierce and Schreibman (1995) to promote social interactions between ASD children and their peers: *gaining attention* (the TD children were taught to attract the attention of the child with ASD before talking to him, offering a choice or giving a demonstration), *providing choices* (the TD children were taught to offer the child with ASD two activity options), *modeling actions* (the TD children were taught to provide frequent and varied examples of appropriate actions, including both task completion and social behaviors), *narrating activities using simple sentences* (the TD were taught to narrate and comment on their own actions on the iPad in language adapted to the level of comprehension of the child with ASD), *turn-taking* (the TD children were taught to encourage the child with ASD to take turns, to wait and to demonstrate sharing by using the iPad concurrently with them), e*ncouraging and extending the conversation* (the TD were taught to initiate and maintain the conversation with the child with ASD, by asking questions about the task or prompting the peer to repeat an appropriate verbal behavior whenever possible), and *reinforcing attempts* (the TD were taught to enthusiastically praise the child with ASD). During the session, the experimenter provided a brief description of each strategy and modeled several examples of its application (see Table 2). A manual containing each of these strategies represented in both written and pictorial form was also used as visual support for the children.

Then, the TD participants reviewed strategies during the third and fourth sessions. Role play was also conducted during these sessions to give the children the opportunity to practice the strategies. Role play was conducted with the experimenter as the student with ASD who needed support to perform activities and interact socially. Feedback was provided at each attempt to use a strategy during role play. When the strategy was used correctly, the TD child was praised by the experimenter. When a strategy was misused, the experimenter provided corrective feedback by remodeling expected behavior and then asking the child to do it again. When a strategy was missed (e.g., did not provide a choice), the experimenter reminded the TD child to use it. Training was considered complete when the TD child could spontaneously and correctly use all strategies.

Intervention

The intervention procedure was identical to that of the baseline phase, except that the experimenter encouraged TD students to use the strategies they had learned. The manual depicting strategies presented during peer training was available, and the TD children could consult it whenever they needed. The experimenter provided the TD participants

Table 2 Specific examples of how strategies were applied

Strategy	Examples
Gaining attention	Tap on the shoulder; Call by first name; "Look what I'm doing"
Providing choices	Show the pictograms of two tasks; "What do you prefer"
Modeling actions	Show where to place the puzzle piece; Show how to do a high-five
Narrating activities using simple sentences	"I'm doing the puzzle"; "I put the yellow with the yellow"; "Wait, I'm preparing the activity"
Turn-taking	Give the tablet; Ask for the tablet; "Your turn now"
Encouraging and extending the conversation	"What is this picture?"; "Show me the car" "Repeat after me"
Reinforcing attempts	Do a high five; "Well done"; "Great, keep going"

with verbal feedback (e.g., “call him first”, “show him how to do it”) and praise (e.g., “It's great what you're doing”). However, he avoided physical guidance during sessions and interacted as little as possible with the child with ASD.

Follow-up

The sessions were conducted immediately after the intervention phase for all participants. During follow-up, conditions were the same as in baseline. The manual was removed from the room and the experimenter ceased to provide verbal feedback. Before each session, the TD participant was instructed to continue to support their peer with ASD, but without assistance from the adult. When the session had started, the experimenter no longer prompted the TD participants to use the learned strategies.

Interobserver Agreement

The first author served as the primary data coder and a trained PhD student in psychology served as reliability coder. The reliability coder was independent of the study, previously trained to collect reliability data, and masked to the experiment's hypotheses. The reliability coder viewed and scored 20% of randomly selected video-taped sessions within each phase and dyad. For all variables, interobserver agreement was calculated using a point-by-point-agreement (Kazdin, 2011). An agreement occurred when both coders recorded an occurrence of the same behavior at the same time and a disagreement occurred when a coder recorded a behavior that was not observed by the other coder. For each variable, an interobserver agreement was then calculated by dividing the number of agreements by the sum of agreements and disagreements and multiplying the result by 100%. On average, interobserver agreement was 88% for initiations, 95% for negative initiations, 82% for responses and 94% for negative responses, demonstrating an acceptable reliability between the two observers.

Treatment Fidelity

The researchers developed a 10-item implementation checklist (available as supplementary material) of core components that were expected during intervention sessions. Specifically, six items concerned the use of teaching strategies (e.g., the TD child offered choices to his peer with ASD), three items concerned the experimenter (e.g., the experimenter praised the TD child), and one item concerned the presence of the manual containing the learning strategies. The first author and a PhD student independently examined 20% of randomly selected intervention sessions for each dyad and scored each item according to its presence or absence in the video. The treatment fidelity was then calculated by dividing the number of items marked as present by the total number of items included in the checklist (i.e., 10) and multiplying by 100%. The average treatment fidelity during the intervention phase reached 97.5% (range 90–100), with an overall agreement of 95% (range 80–100) between the two coders.

Data Analysis

We primarily used visual inspection of individual data to examine the effects of the intervention, according to the single-case design standards (Horner et al., 2005; Krasny-Pacini & Evans, 2018). Specifically, we examined the immediacy, level and pattern of change in the dependent measures compared to baseline. The overlap of data between phases was also considered. To supplement this analysis of all our dependent variables, we calculated Tau-U, a non-parametric effect size metric appropriate for single-case research designs (Parker et al., 2011a). Tau-U calculates the non-overlap of all pairs of data between baseline and intervention while controlling for a trend during baseline when necessary, making it one of the most relevant effect size estimates in single-case research (Parker et al., 2011b). Tau-U trend analysis was conducted for the baseline before conducting Tau-U contrast analysis, but no baseline adjustment was required in the present study as all results were nonsignificant at $p < 0.05$. Tau-U is interpretable as a continuous index of improvement with scores ranging from 0.00 to 1.00. More precisely, a score of 0.65 or below may be considered as a small effect size, between 0.66 and 0.92 as a moderate effect size, and 0.93 and above as a large effect size (Rakap, 2015). We calculated Tau-U with a web-based calculator (https://singlecaseresearch.org/calculators/tau-u/), to provide an estimate of the magnitude of the effect for each participant and for the intervention as a whole. However, despite its advantages, Tau-U has certain limitations, particularly its inability to quantify the magnitude of behavioral change and the lack of consensus on its interpretation using the previously mentioned benchmarks, which should therefore be considered with caution (Vannest & Ninci, 2015).

Results

Initiations

Figure 2 shows the rates of initiations exhibited by participants (with and without ASD) during baseline, intervention, and follow-up phases.

All participants with ASD exhibited stable baselines, with very low rates of initiations during the sessions. After the introduction of the intervention, Bastien increased his rates of initiations slightly from 0.00 to 0.18 behaviors per minute on

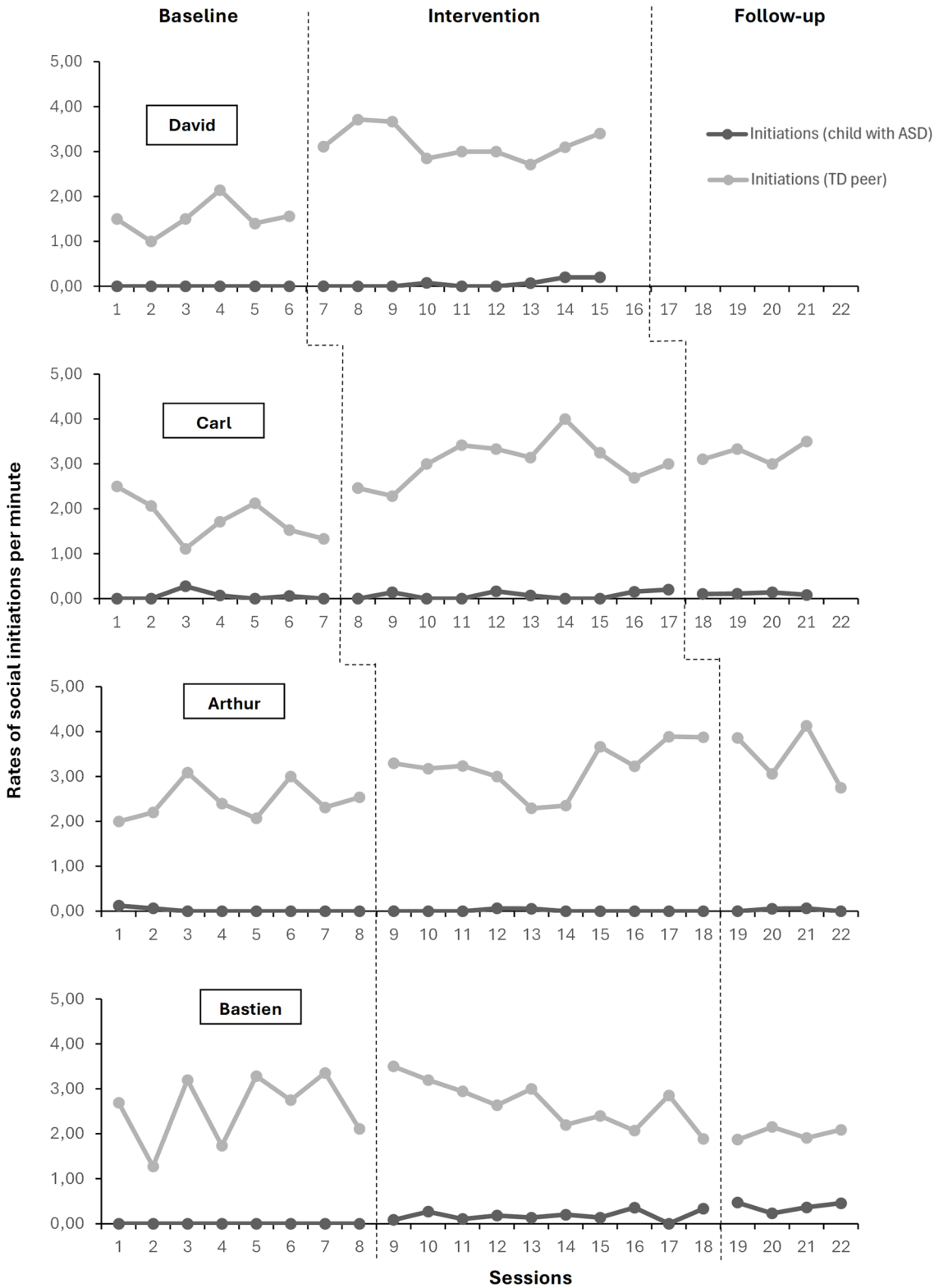


Fig. 2 Rates of initiations exhibited by children with ASD and their TD peers during baseline, intervention and follow-up

average. The change was immediate, with only one data point that overlapped with the baseline. Then, he maintained his gains during follow-up and reached an average of 0.38 behaviors per minute. For David, Carl and Arthur, visual analysis indicated low and stable rates of initiations after the introduction of the intervention, without an observational change in level or trend, and multiple overlapping data points between baseline and intervention. The individual Tau-U analyses confirmed visual analysis and suggested a moderate effect size for Bastien (0.90), and a weak effect size for David (0.44), Carl (0.13) and Arthur (0.09). The weighted average Tau-U for initiations across participants with ASD was 0.35 [95% CI 0.06–0.63], suggesting a weak effect of the intervention.

Among TD children, David's peer exhibited between 1.00 and 2.14 initiations per minute (averaging at 1.52) during baseline, which increased immediately to between 2.71 and 3.71 initiations per minute (averaging at 3.17) after the introduction of the intervention, with no overlap in data between the two conditions. For Carl's peer, initiations varied between 1.11 and 2.50 behaviors per minute during baseline (averaging at 1.7), and then increased in level during the intervention to reach an average of 3.06 initiations per minute (range: 2.29–4.00). The change was immediate after the introduction of the intervention, with only two overlapping data points with baseline, and was then maintained during follow-up (averaging at 3.23 initiations per minute). For Arthur's peer, a similar pattern of change was noted. His overall rate of initiations increased from baseline (mean = 2.45, range 2.00–3.09) to intervention (mean = 3.20, range 2.29–3.89), despite three data points from intervention that overlapped with the baseline phase. Moreover, this change in level was immediate and was maintained during follow-up with an average of 3.45 initiations per minute (range 2.75–4.13). For Bastien's peer, the rate of initiations was between 1.27 and 3.36 behaviors per minute (averaging at 2.55) during baseline. After the introduction of the intervention, we noted a slight decreasing trend until the follow-up phase, although there was no observable change in level (with an average of 2.67 behaviors per minute during intervention) and multiple overlaps between baseline and intervention. The individual Tau-U analyses confirmed visual analysis, suggesting a weak effect size for Bastien's peer (0.04), a moderate effect size for Arthur's peer (0.74), and a strong effect size for both Carl's peer (0.94) and David's peer (1.00). The weighted average Tau-U for initiations across TD participants was 0.67 [95% CI 0.38–0.96], suggesting a moderate effect of the intervention.

Responses

Figure 3 shows the rates of responses exhibited by participants (with and without ASD) during baseline, intervention, and follow-up phases.

All participants with ASD exhibited stable baselines and demonstrated more variability in their social responses after the introduction of the intervention. For David, responses lasted between 0.50 and 1.27 behaviors per minute during baseline (averaging at 0.75), and then increased to reach an average of 3.86 behaviors per minute during the intervention (range: 2.16–5.20), without overlap between these two conditions. The change in level appeared immediately after the introduction of the intervention and the rates of initiations increased progressively throughout the intervention phase. For Carl, rates of responses varied between 1.06 and 2.00 behaviors per minute (averaging at 1.66) before the intervention and then increased to an overall level of 3.12 behaviors per minute (range: 2.31–3.75) after the peer training. The change was immediate, with no overlap between baseline and intervention, and was maintained during follow-up with an average of 3.64 responses per minute (range: 3.00–4.00). A similar pattern of change was also noted for Bastien. During baseline, his rates of responses were between 1.00 and 2.20 behaviors per minute, averaging at 1.77. After the peer training, his rates of responses immediately increased to reach an average of 3.87 behaviors per minute (range: 2.79–4.82), with no overlap between baseline and intervention. Moreover, gains were maintained during follow-up with an average of 3.96 responses per minute (range: 2.92–4.64). For Arthur, we noted an immediate increase in his rates of responses from baseline (mean = 0.52, range: 0.14–1.13) to intervention (mean = 1.54, range: 1.06–2.00), with only one overlapping data point between these two conditions. His gains were then maintained, with an average of 1.94 responses per minute (range: 1.50–2.64) during follow-up. The individual Tau-U analyses confirmed visual analysis, suggesting a strong effect size for all participants (0.98 for Arthur and 1.00 for Bastien, Carl and David). The weighted average Tau-U for responses across participants with ASD was 0.99 [95% CI = 0.71–1.00], suggesting a large effect of the intervention.

Among TD participants, David's peer exhibited responses that lasted between 0.19 and 1.29 behaviors per minute (averaging at 0.72) during baseline, which then increased in level after the peer training to reach 2.04 behaviors per minute (range: 0.74–3.43) during the intervention. The change was not immediate, but there was an increasing trend during the intervention that supported the existence of a functional relationship between intervention and the increase in initiations. For Carl's peer, we noted a stable baseline with low rates of responses (mean = 0.33, range: 0.22–0.44), which then increased during the intervention to reach an average of 1.06 responses per minute (range: 0.33–1.86). The change was immediate with only one data point from the intervention that overlapped the baseline, and was then maintained during follow-up with rates of responses lasting between 1.00 and 1.78

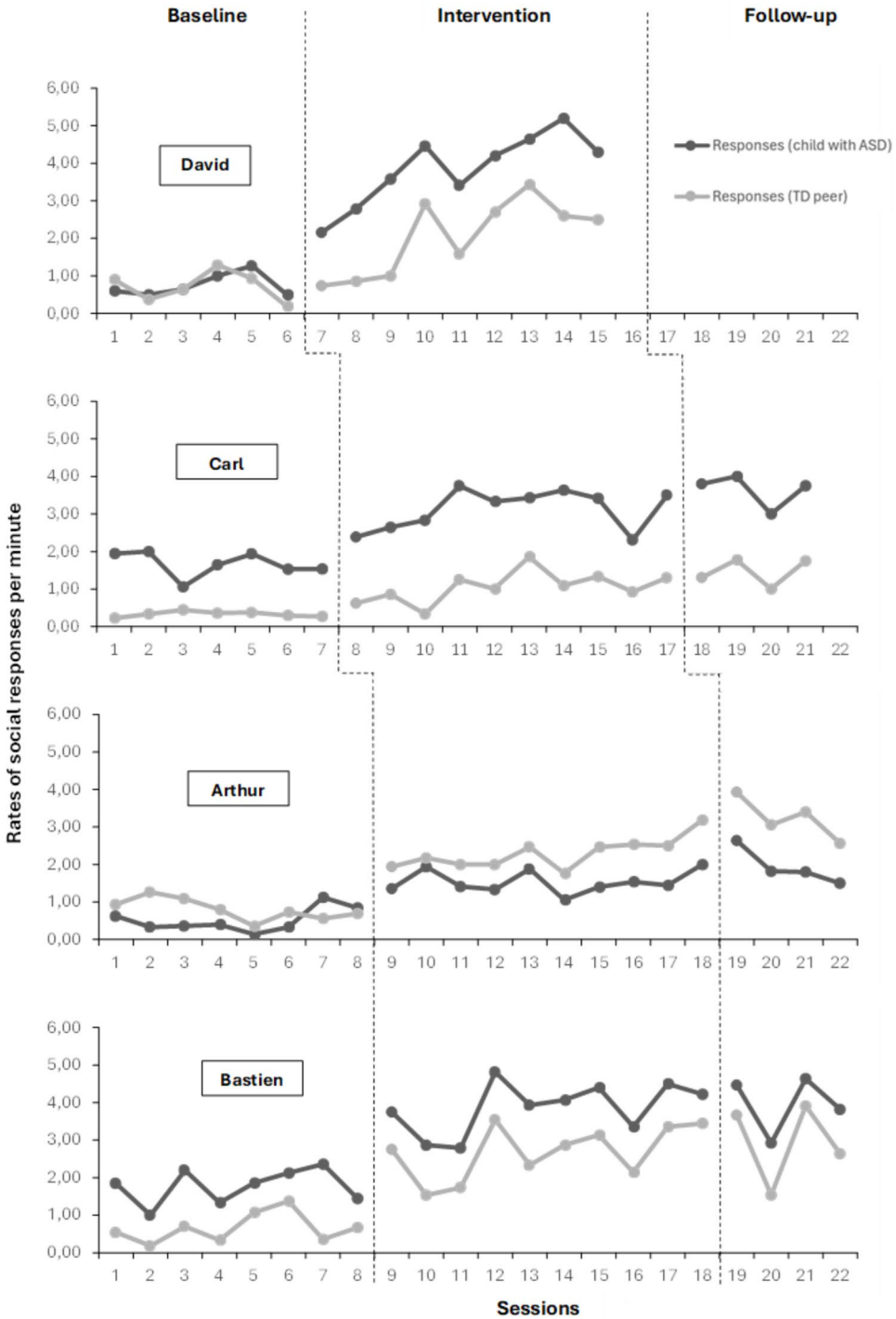


Fig. 3 Rates of responses exhibited by children with ASD and their TD peers during baseline, intervention and follow-up

behaviors per minute (averaging at 1.46). For Arthur's peer, responses lasted between 0.36 and 1.27 behaviors per minute (averaging at 0.81) during baseline. After the peer training, his rates of responses increased to reach an average of 2.30 behaviors per minute (range: 1.76–3.19), with no overlap between baseline and intervention, and then continued to increase during follow-up to reach 3.24 behaviors per minute on average (range: 2.56–3.93). A similar pattern of change was noted for Bastien's peer who exhibited an overall rate of responses during the intervention (mean = 2.68, range: 1.53–1.55) above baseline (mean = 0.65, range: 0.18–1.38), with an immediate change after the introduction of the intervention and no overlap between baseline and intervention. His responses were then maintained during follow-up with 2.94 behaviors per minute on average (range: 1.54–3.91). The

individual Tau-U analyses confirmed the trends observed in the visual analysis and suggested a moderate effect size for Carl's peer (0.90) and David's peer (0.74), and a strong effect size for Arthur's peer (1.00) and Bastien's peer (1.00). The weighted average Tau-U for responses across TD participants was 0.91 [95% CI = 0.63–1.00], suggesting a moderate effect of the intervention.

Negative Initiations and negative responses

Negative initiations and negative responses were never observed in TD children during the study sessions. Consequently, these variables were not plotted for TD participants, and Fig. 4 shows the rates of negative initiations and negative responses exhibited by participants with ASD during baseline, intervention, and follow-up phases.

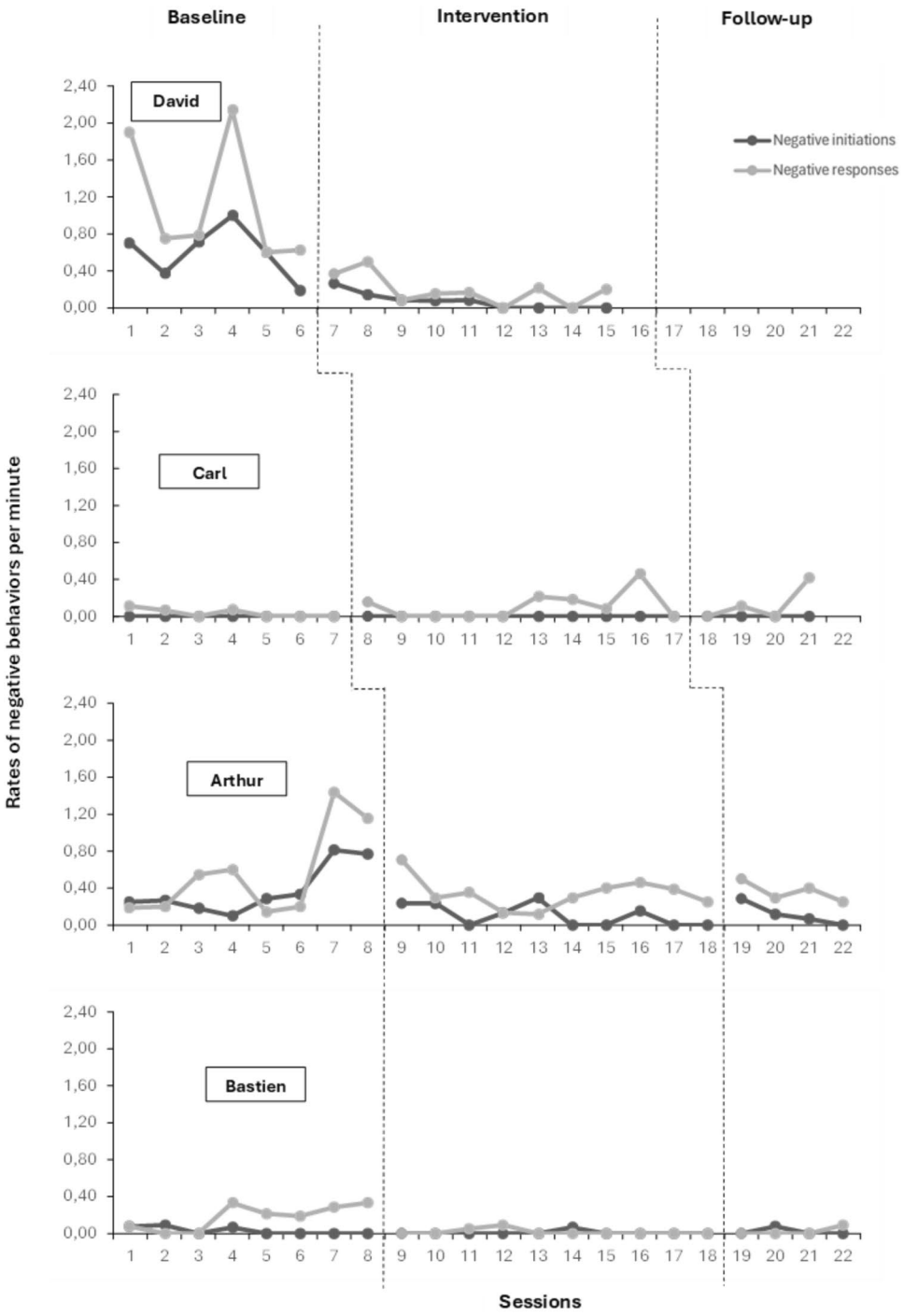


Fig. 4 Rates of negative initiations and negative responses exhibited by children with ASD during baseline, intervention, and follow-up

Concerning negative initiations, David exhibited between 0.19 and 1.00 behaviors per minute (averaging at 0.60) during baseline. After the peer training, his negative initiations tended to decrease throughout the intervention phase (averaging at 0.07 behaviors per minute), until they were no longer observed at all during the last four sessions. Moreover, only one overlapping data point was observed between baseline and intervention. For Arthur, the rate of negative initiations was between 0.10 and 0.81 behaviors per minute (averaging at 0.37) during baseline, and then immediately decreased to reach an average of 0.11 behaviors per minute (range: 0.00–0.29) during the intervention. During follow-up, his rates of negative initiations were maintained at an average of 0.12 behaviors per minute (range: 0.00–0.29). For Carl, negative initiations were never observed during the study sessions. A similar pattern was noted for Bastien who only exhibited negative initiations in three baseline sessions, one intervention session, and one follow-up session. The individual Tau-U analyses confirmed the visual analysis and suggested a strong effect size for David (0.96), a moderate effect size for Arthur (0.74), and a weak effect size for both Carl (0.00) and Bastien (0.30). The weighted average Tau-U for negative initiations across participants with ASD was 0.49 [95% CI = 0.21–0.78], suggesting a weak effect of the intervention.

Concerning negative responses, David exhibited between 0.60 and 2.14 behaviors per minute during baseline (averaging at 1.13), which immediately decreased after the peer training to reach an average of 0.19 behaviors per minute during the intervention (range: 0.00–0.50), without overlap in data between these two conditions. For Carl, negative responses were rarely observed throughout the study sessions. Specifically, he exhibited negative responses in three baseline sessions (mean = 0.04 behaviors per minute, range: 0.07–0.11) and five intervention sessions (mean = 0.11 behaviors per minute, range: 0.08–0.46), with multiple overlapping data points between the two phases. For Arthur, visual analysis indicated between 0.14 and 1.44 negative responses per minute (averaging at 0.56) during baseline, and then between 0.12 and 0.71 negative responses per minute (averaging at 0.34) during the intervention, with significant overlaps between the two conditions. Finally, Bastien exhibited between 0.00 and 0.33 negative responses per minute (averaging at 0.18) in the baseline. After the introduction of the intervention, his rates of negative responses immediately decreased to reach an average of 0.01 behaviors per minute (range: 0.00–0.009), and were then maintained during follow-up with an average of 0.02 behaviors per minute (range: 0.00–0.09). The individual Tau-U analyses support visual analyses and suggested a strong effect size for David (1.00), a moderate effect size for Bastien (0.68), and a weak effect size for both Carl (0.26) and Arthur (0.15). The weighted average Tau-U for negative initiations across participants with ASD was 0.38 [95% CI 0.10–0.67], suggesting a weak effect of the intervention.

Discussion

The purpose of this study was to evaluate the effects of a PM-PRT during iPad-based academic tasks on the social skills of four students with ASD and their TD peers. First, the results showed positive effects of PM-PRT on some social behaviors of the four participants with ASD and their peers, although some variability was noted. After the introduction of the intervention, we observed that three of the four TD children initiated more social interactions, that all participants with ASD increased their social responses, and that all TD peers also increased their rates of social responses. In other words, the intervention led children with ASD to become more socially active by promoting more appropriate social behaviors toward their TD peers, which is consistent with previous studies about PM-PRT (Boudreau et al., 2021; Brock et al., 2018; Harper et al., 2008; Kuhn et al., 2008; Pierce & Schreibman, 1995). The current study also highlighted that the PM-PRT procedures can be successfully implemented in an academic context, thus extending previous literature which was essentially based on recreational situations, and possibly giving school workers greater flexibility in its application. Thus, our findings suggest that PM-PRT may be a promising and adaptable approach for improving the social participation of students with ASD at school, although further investigations are required before it can be designed as an evidence-based practice (Boudreau et al., 2021).

However, despite the encouraging findings that we just described, the effect of the intervention on social initiation by the children with ASD was minimal in the present study. Specifically, only one of the four participants with ASD improved (moderately) in this respect. We have considered two possible explanations for this result. First, the nature of the treatment procedures may be incriminated. Indeed, strategies taught to the TD children focus essentially on how to initiate and maintain positive exchanges, and less on how to create a context enabling children with ASD to spontaneously initiate interactions. This explanation is consistent with the results of Boudreau et al. (2021), which indicated an absence of functional relation between PM-PRT and change in initiations. In addition, on a larger scale, differences between the effects observed on social initiations and those observed on social responses are common among children with ASD who have taken part in a PMI. For example, Martinez et al. (2021) reviewed 18 studies including 53 children with ASD and concluded that PMIs are effective for increasing interactions between participants but may not be as effective for increasing social initiations made by children

with ASD. Together, these findings highlighted the need to think about how to promote social initiation in children with ASD within PMI procedures, due to its importance to developing positive relationships with peers. To this end, incorporating preferred stimuli or video modeling into PMI studies has shown promising results (Dueñas et al., 2021; Thiemann-Bourque et al., 2017), and may require further research. Another explanation based on children's abilities could be advanced to explain our results. Interestingly, the only participant with ASD who responded to PM-PRT on social initiations (i.e., Bastien) was also the only child who used language to communicate, and all his initiations were produced verbally. Language is an important resource whose absence may make it more difficult for some children with ASD to take appropriate initiatives. Thus, future studies about PM-PRT including minimally verbal children may also benefit from training TD children to use augmentative and alternative communication tools (e.g., Thiemann-Bourque et al., 2018). Finally, it is possible that academic activities may not provide the best setting for children with ASD to learn social initiations, and that a more motivating and socially engaging context may be more suitable.

Second, the present study also suggested that PM-PRT during tablet-based academic tasks may decrease inappropriate social behaviors in some students with ASD. Specifically, all participants who exhibited negative initiations and two of the three participants who exhibited negative responses during baseline improved in these domains after the introduction of the intervention. These data support prior literature showing that PMI may contribute to reducing challenging behaviors and inappropriate communication acts among individuals with ASD (Chung et al., 2007; Lee et al., 2007; Thomas & Bambara, 2020). Such findings can be linked to a core characteristic of PMI which is to teach TD students to adapt their communication strategies to the needs of their peers with ASD (Sperry et al., 2010), thereby reducing disruptive behaviors resulting from social misunderstandings. Yet, we also noticed that negative behaviors exhibited by participants with ASD during baseline sessions were sometimes due to the frustration of waiting for one's turn to use a support as pleasurable as the iPad. Although these anecdotal observations should be treated with caution and call for an analysis of communication functions (Tsamitrou & Plumet, 2024), such reports may suggest that incorporating peer-mediated strategies into technology-based instructions may limit some maladaptive behaviors in the case of digital shared workspaces.

Finally, the results demonstrated that the improvements observed during the intervention were maintained when the adult support was withdrawn. This was true for all participants in each dyad for whom maintenance data were collected. The maintenance observed in the present study is in accordance with several previous PMI studies (e.g., Katz & Girolametto, 2015; McFadden et al., 2014), as well as with the only two previous studies that examined maintenance gains in the specific context of PM-PRT (Boudreau et al., 2021; Pierce & Schreibman, 1995). In the study by Pierce and Schreibman (1995), both participants with ASD maintained prolonged interactions, social initiations, and joint attention behaviors at two months follow-up. Similarly, Boudreau et al. (2021) observed that the increased rates of peer engagement and social initiation of two students with ASD were maintained at 6 to 9 weeks following training. The present study contributes to the existing literature by assessing the social behaviors of all participants, not only those with ASD, and demonstrates that TD children also continued to initiate and maintain social interactions with their peers with ASD after the intervention. Such findings are promising and potentially important for practice because they suggest that the positive exchanges developed between students through the PM-PRT may be sustained in more ecological contexts where constant support and prompting from an adult facilitator is rarely possible.

Limitations and Future Directions

There are several limitations of the present study. First, there is a lack of direct assessment of social validity and generalization. Although anecdotal exchanges with TD peers suggested that they enjoyed the process, future research should adopt a more rigorous approach to assess children's acceptance of a tablet-based PM-PRT and evaluate its social validity across core dimensions such as the social significance of treatment goals, the social appropriateness of procedures or the social importance of observed effects (Ferguson et al., 2018). Future studies should also assess the generalization of effects to examine interactions between participants beyond the study sessions. Considering the positive changes observed in the behaviors of TD children, future research could also investigate the potential of PMI-PRT to enhance social acceptance and reduce the stigmatization of children with ASD. Second, the number and the type of academic tasks for each dyad did not vary during data collection across all conditions. Although this was set up to increase the internal validity of the study, future studies evaluating the effects of PM-PRT under academic conditions that are closer to the "real world" would be beneficial. In addition, we cannot rule out the possibility that the nature and difficulty of each task, which differed from one child to another, may have influenced outcomes. Third, the effect of the intervention on the academic achievement of children with ASD has not been assessed, despite the positive effect that the use of iPad technology can have on this domain (Aspiranti et al., 2020). Thus, including broader outcome measures in future research could shed light on whether peer-mediated strategies implemented in academic contexts can provide

a significant effect on both social and academic outcomes. Finally, tablet-based instruction was not experimentally manipulated in the current study; therefore, it remains unclear whether technology-based PM-PRT procedures or traditional medium-based PM-PRT procedures yield different results at the level of student outcomes. Given that digital tablets become increasingly commonplace in school settings (Falloon, 2023), it would be useful for teachers and school workers to know whether using an iPad rather than paper and pencil adds significant benefits in the context of PM-PRT.

Conclusion

The present study indicated that the PM-PRT can be implemented in mainstream school settings to address the social needs of children with ASD. Benefits were noted for all participants, although the intervention was more effective in increasing social responses than social initiations. Additionally, maintenance of the improvements was systematic. Overall, these findings suggested that a PMI adapted from the PM-PRT model and conducted during iPad-based academic tasks may be a viable option for addressing the inclusive education of students with ASD, especially for promoting positive social interactions with TD peers.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10803-025-06920-x.

Acknowledgments The authors would like to thank all the children and their parents who agreed to take part in this study. The authors also thank the school staff for their help and contributions in organizing the sessions. Finally, the authors would like to thank the Rotary Club of Vannes for funding the iPad and Gaele Regnault, for providing access to the Preschool® application.

Author contribution SLS-B acquired funding and supervised the project. GB, SLS-B and GLM-I designed and conceived the study. GB collected, analyzed and interpreted the data. TS helped recruit the participants. GB drafted the manuscript and SLSB, GLM-I and TS revised it critically. All authors read and approved the final manuscript.

Funding This work was supported by the Caisse nationale de la solidarité et de l'autonomie (Iresp-17-AUT4-06) and the Region Bretagne (ARED 2015-2018), ADAPEI 56 (ARED 2015-2018).

Declarations

Conflict of interest The authors declare that they have no competing interests.

References

Alzrayer, N. M., Banda, D. R., & Koul, R. K. (2019). The effects of systematic instruction in teaching multistep social-communication skills to children with autism spectrum disorder using an iPad. *Developmental Neurorehabilitation, 22*(6), 415–429. https://doi.org/10.1080/17518423.2019.1604578

Aspiranti, K. B., Larwin, K. H., & Schade, B. P. (2020). iPads/tablets and students with autism: A meta-analysis of academic effects. *Assistive Technology, 32*(1), 23–30. https://doi.org/10.1080/10400435.2018.1463575

Aubé, B., Follenfant, A., Goudeau, S., & Derguy, C. (2021). Public stigma of autism spectrum disorder at school: Implicit attitudes matter. *Journal of Autism and Developmental Disorders, 51*(5), 1584–1597. https://doi.org/10.1007/s10803-020-04635-9

Barber, A. B., Saffo, R. W., Gilpin, A. T., Craft, L. D., & Goldstein, H. (2016). Peers as clinicians: Examining the impact of Stay Play Talk on social communication in young preschoolers with autism. *Journal of Communication Disorders, 59*, 1–15. https://doi.org/10.1016/j.jcomdis.2015.06.009

Beckman, A., Mason, B. A., Wills, H. P., Garrison-Kane, L., & Huffman, J. (2019). Improving behavioral and academic outcomes for students with autism spectrum disorder: Testing an app-based self-monitoring intervention. *Education and Treatment of Children, 42*(2), 225–244. https://doi.org/10.1353/etc.2019.0011

Bondy, A. S., & Frost, L. A. (1994). The picture exchange communication system. *Focus on Autistic Behavior, 9*(3), 1–19. https://doi.org/10.1177/108835769400900301

Boudreau, A. M., Corkum, P., Meko, K., & Smith, I. M. (2015). Peer-mediated pivotal response treatment for young children with autism spectrum disorders: A systematic review. *Canadian Journal of School Psychology, 30*(3), 218–235. https://doi.org/10.1177/0829573515581156

Boudreau, A. M., Lucyshyn, J. M., Corkum, P., Meko, K., & Smith, I. M. (2021). Peer-mediated pivotal response treatment at school for children with autism spectrum disorder. *Canadian Journal of School Psychology, 36*(1), 34–50. https://doi.org/10.1177/0829573519870923

Bourgueil, O., Regnault, G., & Moutier, S. (2015). Création d'outils numériques pour personnes avec trouble du spectre de l'autisme: De la recherche à la pratique, et vice versa. *Enfance, 1*(1), 11–126. https://doi.org/10.4074/S001375451500107X

Briet, G., Le Maner-Idrissi, G., Seveno, T., Lemarec, O., & Le Sourn-Bissaoui, S. (2021). Developmental follow-up of children with autism spectrum disorder enrolled in inclusive units in france: Outcomes and correlates of change. *International Journal of Disability, Development and Education*. https://doi.org/10.1080/1034912X.2021.1921123

Briet, G., Maner-Idrissi, G. L., Seveno, T., Marec, O. L., & Sourn-Bissaoui, S. L. (2023). Peer mediation in play settings for minimally verbal students with autism spectrum disorder. *Autism and Developmental Language Impairments, 8*, 1–17. https://doi.org/10.1177/23969415231204837

Brock, M. E., Dueker, S. A., & Barczak, M. A. (2018). Brief report: Improving social outcomes for students with autism at recess through peer-mediated pivotal response training. *Journal of Autism and Developmental Disorders, 48*(6), 2224–2230. https://doi.org/10.1007/s10803-017-3435-3

Bruner, J. (1983). *Le développement de l'enfant. Savoir faire, savoir dire*. PUF.

Camargo, S. P. H., Rispoli, M., Ganz, J., Hong, E. R., Davis, H., & Mason, R. (2016). Behaviorally based interventions for teaching social interaction skills to children with ASD in inclusive settings: A meta-analysis. *Journal of Behavioral Education, 25*(2), 223–248. https://doi.org/10.1007/s10864-015-9240-1

Charlop, M. H., Lang, R., & Rispoli, M. (2018). Want to play? Peer-mediated intervention for young children with autism spectrum disorder. In M. H. Charlop, R. Lang, & M. Rispoli (Eds.), *Play and social skills for children with autism spectrum disorder* (pp. 107–127). Springer. https://doi.org/10.1007/978-3-319-72500-0_7

Christ, T. J. (2007). Experimental control and threats to internal validity of concurrent and nonconcurrent multiple baseline designs. *Psychology in the Schools, 44*(5), 451–459. https://doi.org/10.1002/pits.20237

Chung, K.-M., Reavis, S., Mosconi, M., Drewry, J., Matthews, T., & Tassé, M. J. (2007). Peer-mediated social skills training program for young children with high-functioning autism. *Research in Developmental Disabilities, 28*(4), 423–436. https://doi.org/10.1016/j.ridd.2006.05.002

Crosland, K., & Dunlap, G. (2012). Effective strategies for the inclusion of children with autism in general education classrooms. *Behavior Modification, 36*(3), 251–269. https://doi.org/10.1177/0145445512442682

Dueñas, A. D., Plavnick, J. B., & Goldstein, H. (2021). Effects of a multicomponent peer mediated intervention on social communication of preschoolers with autism spectrum disorder. *Exceptional Children, 87*(2), 236–257. https://doi.org/10.1177/0014402920953761

Fage, C., Consel, C. Y., Balland, E., Etchegoyhen, K., Amestoy, A., Bouvard, M., & Sauzéon, H. (2018). Tablet apps to support first school inclusion of children with Autism Spectrum Disorders (ASD) in mainstream classrooms: A pilot study. *Frontiers in Psychology, 9*, 1–16. https://doi.org/10.3389/fpsyg.2018.02020

Falloon, G. (2023). Twelve years of iPads and apps in schools: What conditions support effective practices in K-6 classrooms? *Frontiers in Education, 8*, 1–12. https://doi.org/10.3389/feduc.2023.1122522

Ferguson, J. L., Cihon, J. H., Leaf, J. B., Van Meter, S. M., McEachin, J., & Leaf, R. (2018). Assessment of social validity trends in the journal of applied behavior analysis. *European Journal of Behavior Analysis, 20*(1), 146–157. https://doi.org/10.1080/15021149.2018.1534771

Gal, E., Lamash, L., Bauminger-Zviely, N., Zancanaro, M., & Weiss, P. L. (2016). Using multitouch collaboration technology to enhance social interaction of children with high-functioning autism. *Physical and Occupational Therapy in Pediatrics, 36*(1), 46–58. https://doi.org/10.3109/01942638.2015.1040572

Guillemot, F., Lacroix, F., & Nocus, I. (2022). Teachers' attitude towards inclusive education from 2000 to 2020: An extended meta-analysis. *International Journal of Educational Research Open, 3*, Article 100175. https://doi.org/10.1016/j.ijedro.2022.100175

Gunning, C., Breathnach, Ó., Holloway, J., McTiernan, A., & Malone, B. (2019). A systematic review of peer-mediated interventions for preschool children with autism spectrum disorder in inclusive settings. *Review Journal of Autism and Developmental Disorders, 6*(1), 40–62. https://doi.org/10.1007/s40489-018-0153-5

Haas, A., Vannest, K., & Smith, S. D. (2019). Utilizing peers to support academic learning for children with Autism Spectrum Disorder. *Behavior Analysis in Practice, 12*, 734–740. https://doi.org/10.1007/s40617-019-00363-4

Hansen, S. G., Mowbray, M., Raulston, T., Carnett, A., & Tullis, C. (2023). Effects of a peer-mediated joint attention intervention in an inclusive preschool setting. *Focus on Autism and Other Developmental Disabilities, 38*(2), 71–79. https://doi.org/10.1177/10883576221108111

Harper, C. B., Symon, J. B. G., & Frea, W. D. (2008). Recess is time-in: Using peers to improve social skills of children with autism. *Journal of Autism and Developmental Disorders, 38*(5), 815–826. https://doi.org/10.1007/s10803-007-0449-2

Horner, R. H., Carr, E. G., Halle, J., McGee, G., Odom, S., & Wolery, M. (2005). The use of single-subject research to identify evidence-based practice in special education. *Exceptional Children, 71*(2), 165–179. https://doi.org/10.1177/001440290507100203

Hu, X., Zheng, Q., & Lee, G. T. (2018). Using peer-mediated LEGO® Play intervention to improve social interactions for chinese children with autism in an inclusive setting. *Journal of Autism and Developmental Disorders, 48*(7), 2444–2457. https://doi.org/10.1007/s10803-018-3502-4

Kamps, D., Barbetta, P. M., Leonard, B. R., & Delquadri, J. (1994). Classwide peer tutoring: An integration strategy to improve reading skills and promote peer interactions among students with autism and general education peers. *Journal of Applied Behavior Analysis, 27*(1), 49–61. https://doi.org/10.1901/jaba.1994.27-49

Katz, E., & Girolametto, L. (2015). Peer-mediated intervention for preschoolers with ASD: Effects on responses and initiations. *International Journal of Speech-Language Pathology, 17*(6), 565–576. https://doi.org/10.3109/17549507.2015.1024166

Kazdin, A. E. (2011). *Single-case research designs: Methods for clinical and applied setting* (2nd ed.). Oxford University Press.

Kent, C., Cordier, R., Joosten, A., Wilkes-Gillan, S., & Bundy, A. (2021). Can I learn to play? Randomized control trial to assess effectiveness of a peer-mediated intervention to improve play in children with autism spectrum disorder. *Journal of Autism and Developmental Disorders, 51*(6), 1823–1838. https://doi.org/10.1007/s10803-020-04671-5

Koegel, L. K., Koegel, R. L., Harrower, J. K., & Carter, C. M. (1999). Pivotal response intervention: Overview of approach. *Research and Practice for Persons with Severe Disabilities, 24*(3), 174–185. https://doi.org/10.2511/rpsd.24.3.174

Koegel, R. L., Koegel, L. K., & McNerney, E. K. (2001). Pivotal areas in intervention for autism. *Journal of Clinical Child and Adolescent Psychology, 30*(1), 19–32. https://doi.org/10.1207/S15374424JCCP3001_4

Krasny-Pacini, A., & Evans, J. (2018). Single-case experimental designs to assess intervention effectiveness in rehabilitation: A practical guide. *Annals of Physical and Rehabilitation Medicine, 61*(3), 164–179. https://doi.org/10.1016/j.rehab.2017.12.002

Kuhn, L. R., Bodkin, A. E., Devlin, S. D., & Doggett, R. A. (2008). Using pivotal response training with peers in special education to facilitate play in two children with autism. *Education and Training in Developmental Disabilities, 43*(1), 37–45.

Kurth, J. A., Miller, A. L., Toews, S. G., Thompson, J. R., Cortés, M., Dahal, M. H., De Escallón, I. E., Hunt, P. F., Porter, G., Richler, D., Fonseca, I., Singh, R., Šiška, J., Villamero, R., Jr., & Wangare, F. (2018). Inclusive education: Perspectives on implementation and practice from international experts. *Intellectual and Developmental Disabilities, 56*(6), 471–485. https://doi.org/10.1352/1934-9556-56.6.471

Lee, S., Odom, S. L., & Loftin, R. (2007). Social engagement with peers and stereotypic behavior of children with autism. *Journal of Positive Behavior Interventions, 9*(2), 67–79. https://doi.org/10.1177/10983007070090020401

Lord, C., Rutter, M., DiLavore, P. C., Risi, S., Gotham, K., & Bishop, S. (2012). *Autism diagnostic observation schedule: ADOS-2*. Western Psychological Services.

Martinez, J. R., Waters, C. L., Conroy, M. A., & Reichow, B. (2021). Peer-mediated interventions to address social competence needs of young children with ASD: Systematic review of single-case research design studies. *Topics in Early Childhood Special Education, 40*(4), 217–228. https://doi.org/10.1177/0271121419839136

McFadden, B., Kamps, D., & Heitzman-Powell, L. (2014). Social communication effects of peer-mediated recess intervention for children with autism. *Research in Autism Spectrum Disorders, 8*(12), 1699–1712. https://doi.org/10.1016/j.rasd.2014.08.015

Moeyaert, M., Klingbeil, D. A., Rodabaugh, E., & Turan, M. (2021). Three-level meta-analysis of single-case data regarding the effects of peer tutoring on academic and social-behavioral outcomes for at-risk students and students with disabilities. *Remedial and Special Education, 42*(2), 94–106. https://doi.org/10.1177/0741932519855079

Neely, L., Rispoli, M., Camargo, S., Davis, H., & Boles, M. (2013). The effect of instructional use of an iPad® on challenging behavior and academic engagement for two students with autism. *Research in Autism Spectrum Disorders, 7*(4), 509–516. https://doi.org/10.1016/j.rasd.2012.12.004

Odom, S. L., & Strain, P. S. (1984). Peer-mediated approaches to promoting children's social interaction: A review. *American Journal of Orthopsychiatry, 54*(5), 544–557. https://doi.org/10.1111/j.1939-0025.1984.tb01525.x

Paquet, A., Meilhoc, L., Mas, B., Morena, A.-S., & Girard, M. (2022). Does playing on a digital tablet impact the social interactions of children with autism spectrum disorder? *Neuropsychiatrie De L'enfance Et De L'adolescence, 70*(6), 296–307. https://doi.org/10.1016/j.neurenf.2022.04.003

Park, I., Gong, J., Lyons, G. L., Hirota, T., Takahashi, M., Kim, B., Lee, S., Kim, Y. S., Lee, J., & Leventhal, B. L. (2020). Prevalence of and factors associated with school bullying in students with autism spectrum disorder: A cross-cultural meta-analysis. *Yonsei Medical Journal, 61*(11), 909. https://doi.org/10.3349/ymj.2020.61.11.909

Parker, R., Vannest, K., & Davis, J. (2011a). Effect size in single-case research: A review of nine nonoverlap techniques. *Behavior Modification, 35*(4), 303–322. https://doi.org/10.1177/0145445511399147

Parker, R., Vannest, K., Davis, J., & Sauber, S. (2011b). Combining nonoverlap and trend for single-case research: Tau-U. *Behavior Therapy, 42*(2), 284–299. https://doi.org/10.1016/j.beth.2010.08.006

Pierce, K., & Schreibman, L. (1995). Increasing complex social behaviors in children with autism: Effects of peer-implemented pivotal response training. *Journal of Applied Behavior Analysis, 28*(3), 285–295. https://doi.org/10.1901/jaba.1995.28-285

Rakap, S. (2015). Effect sizes as result interpretation aids in single-subject experimental research: Description and application of four nonoverlap methods. *British Journal of Special Education, 42*(1), 11–33. https://doi.org/10.1111/1467-8578.12091

Rosenbloom, R., Wills, H. P., Mason, R., Huffman, J. M., & Mason, B. A. (2019). The effects of a technology-based self-monitoring intervention on on-task, disruptive, and task-completion behaviors for adolescents with autism. *Journal of Autism and Developmental Disorders, 49*(12), 5047–5062. https://doi.org/10.1007/s10803-019-04209-4

Rutter, M., Le Couteur, A., & Lord, C. (2003). *Autism diagnostic interview-revised*. Western Psychological Services.

Schopler, E., Lansing, M. D., Reichler, M. J., & Marcus, L. M. (2005). *Psychoeducational profile-third edition (PEP-3)*. Western Psychological Services.

Sperry, L., Neitzel, J., & Engelhardt-Wells, K. (2010). Peer-mediated instruction and intervention strategies for students with autism spectrum disorders. *Preventing School Failure: Alternative Education for Children and Youth, 54*(4), 256–264. https://doi.org/10.1080/10459881003800529

Tan, P., & Alant, E. (2016). Using peer-mediated instruction to support communication involving a student with autism during mathematics activities: A case study. *Assistive Technology, 30*(1), 9–15. https://doi.org/10.1080/10400435.2016.1223209

Tellegen, P., Laros, J., & Kiat, M. (2009). *SON-R 21/2—7—Test non-verbal intelligence. Adaptation Française*. Hogrefe.

Thiemann-Bourque, K., Feldmiller, S., Hoffman, L., & Johner, S. (2018). Incorporating a peer-mediated approach into speech-generating device intervention: Effects on communication of preschoolers with autism spectrum disorder. *Journal of Speech, Language, and Hearing Research, 61*(8), 2045–2061. https://doi.org/10.1044/2018_JSLHR-L-17-0424

Thiemann-Bourque, K. S., McGuff, S., & Goldstein, H. (2017). Training peer partners to use a speech-generating device with classmates with autism spectrum disorder: Exploring communication outcomes across preschool contexts. *Journal of Speech, Language, and Hearing Research, 60*(9), 2648–2662. https://doi.org/10.1044/2017_JSLHR-L-17-0049

Thomas, A., & Bambara, L. M. (2020). Using peer-mediation to enhance conversation and reduce inappropriate communication acts in adolescents with autism. *Education and Training in Autism and Developmental Disabilities, 55*(2), 185–200.

Tsamitrou, S., & Plumet, M.-H. (2024). The importance and challenges of observing social interactions in autistic preschoolers during inclusive educational settings: A scoping review. *Autism and Developmental Language Impairments, 9*, 23969415241227076. https://doi.org/10.1177/23969415241227077

Vannest, K. J., & Ninci, J. (2015). Evaluating intervention effects in single-case research designs. *Journal of Counseling and Development, 93*(4), 403–411. https://doi.org/10.1002/jcad.12038

Vygotsky, L. S. (1978). *Mind in society*. Harvard University Press.

Watkins, L., O'Reilly, M., Kuhn, M., Gevarter, C., Lancioni, G. E., Sigafoos, J., & Lang, R. (2015). A review of peer-mediated social interaction interventions for students with autism in inclusive settings. *Journal of Autism and Developmental Disorders, 45*(4), 1070–1083. https://doi.org/10.1007/s10803-014-2264-x

Watson, P. J., & Workman, E. A. (1981). The non-concurrent multiple baseline across-individuals design: An extension of the traditional multiple baseline design. *Journal of Behavior Therapy and Experimental Psychiatry, 12*(3), 257–259. https://doi.org/10.1016/0005-7916(81)90055-0

Whalon, K. J., Conroy, M. A., Martinez, J. R., & Werch, B. L. (2015). School-based peer-related social competence interventions for children with autism spectrum disorder: A meta-analysis and descriptive review of single case research design studies. *Journal of Autism and Developmental Disorders, 45*(6), 1513–1531. https://doi.org/10.1007/s10803-015-2373-1

Wolfberg, P., Bottema-Beutel, K., & DeWitt, M. (2012). Including children with autism in social and imaginary play with typical peers: integrated play groups model. *American Journal of Play, 5*(1), 55–80.

Zhang, B., Liang, S., Chen, J., Chen, L., Chen, W., Tu, S., Hu, L., Jin, H., & Chu, L. (2022). Effectiveness of peer-mediated intervention on social skills for children with autism spectrum disorder: A randomized controlled trial. *Translational Pediatrics, 11*(5), 663–675. https://doi.org/10.21037/tp-22-110

Zhang, J., & Wheeler, J. J. (2011). A meta-analysis of peer-mediated interventions for young children with autism spectrum disorders. *Education and Training in Autism and Developmental Disabilities, 46*(1), 62–77.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.